

Pre-Reads: Loops & Flow Control



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Pre-Reads

Foundation

Recommended

Resource



Associated Lectures (1)



Description



Discussions



Pre-Read: Loops & Flow Control in Python



Why This Session Matters

If variables are the memory of a computer, **loops are its superpower**. They allow a computer to repeat actions thousands of times without getting tired, distracted, or bored.

Imagine having to write:

```
print(1) print(2) print(3) ... all the way to 1000.
```

You'd go crazy. Python wouldn't.

Loops turn long, repetitive work into short, efficient instructions. This is one of the big reasons Python feels more powerful than tools like Excel where you drag formulas down endlessly.

In this session, you'll finally understand how computers think when they repeat tasks.



The Big Idea Behind Loops

Think of loops like a **smart assistant** that keeps doing a task until you tell it to stop.

A few real-life examples:

- A washing machine spinning **until** the timer ends
- A playlist repeating **for each** song in your liked list
- Google Photos checking **every** image for faces
- A fitness app giving reminders **while** you haven't completed your steps

Loops in Python work exactly like these real-world patterns.



Two Main Types of Loops

1. The “keep doing this while it stays true” loop

This is like: "Keep stirring the soup while it's still lumpy." You stop only when the condition changes.

Important idea: If you never stop stirring, the soup burns. If you never stop a loop, your program hangs. That's why learning how to avoid infinite loops is important.

2. The “do this for every item” loop

This is like: "Give chocolates to every student in the class." You know the group, and you want to repeat an action for each one.

The range function helps you go through numbers easily, like moving through a predictable list.



Loop Direction Controls

Inside loops, you sometimes need to change the flow.

Think of it like walking down a road:

- **break** You stop walking immediately and turn around.
 - **continue** You skip one step and jump to the next one.
 - **pass** You keep walking as if nothing happened. It's like saying "I'll handle this part later."
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A Fun Challenge Coming Up

You'll practice loops through a famous brain teaser called **FizzBuzz**. It looks simple at first, but it teaches you how loops, conditions, and logical thinking come together. This is a favorite warm-up question in tech interviews because it shows how well you think, not how much you know.



Before the Session

Think about moments in your life where something repeats automatically:

- Alarms
- Billing cycles
- Daily streaks in apps
- Timers
- Fitness goals
- Traffic signals

These are real-world loops. Python loops work the same way, just in a more controlled and flexible manner.

Bring that curiosity to class. Once you understand loops, you'll start seeing them everywhere in the digital world.

✨ Closing Thought

“Loops let computers do in seconds what would take humans hours.”

This session will help you think like a programmer: spot patterns, reduce repetition, and write cleaner, smarter instructions.
